SECURITY PROPOSAL FOR MANAGEMENT · OPTION COMPARISON · GENERAL EDITION

Proposal 2 vs Proposal 3

■ **Proposal 2, AWS edge:** Amazon CloudFront with AWS WAF at the edge in front of the origin load balancer. ■ **Proposal 3, existing edge:** the Imperva Cloud WAF site stays as the first filter, with AWS WAF at the origin. Both keep the same administrator path: AWS Verified Access, IAM Identity Center with mandatory MFA, an administrator web ACL and CloudShell for break-glass.

Basis: the management proposal items p2-* and p3-*, the strategy deep dives 8.1 and 8.7, the budget worksheet defaults (10 million public requests a month, 730 hours, up to 100 administrators), us-east-1 data transfer prices, and official AWS and Imperva documentation re-checked on 29 October 2026. Amounts are USD list prices or the worksheet's planning placeholders, never vendor quotes.

Home Region us-east-1

One administrator path in both options

Independent PDF and Markdown

Supporting evidence opens in a new tab. Internal citations and exports refer to this GitHub-deployed edition; external documentation remains unchanged.

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01 · AT A GLANCE

The decision in one table

Factor	Proposal 2 · AWS edge	Proposal 3 · Existing edge (Imperva)
Public edge	One CloudFront distribution on a flat-rate plan, with an AWS WAF web ACL, Shield Standard and the Anti-DDoS managed rules that the Business and Premium plans bundle. CloudFront plans	The existing Imperva Cloud WAF site: CDN, DDoS mitigation, WAF rules policy, bot access control and load balancing in one service, on the tier the contract specifies (open decision B22). Imperva datasheet
Who sees the real visitor	CloudFront and the edge web ACL see the client address, so the AWS IP reputation and anonymous IP lists work at the edge. The origin reads the client from X-Forwarded-For. AWS WAF IP reputation	Only Imperva sees the client. The origin sees Imperva addresses and reads the visitor from the Incap-Client-IP header, so the AWS reputation lists stay off at the origin and Imperva's own reputation features are used. Imperva checklist
Where the public WAF runs	Twice in AWS WAF: the edge web ACL on CloudFront (managed rules, route allow-list, rate rules, country labels) and the origin web ACL in BLOCK as a second layer.	Two engines: Imperva's WAF Rules policy, ACL and custom rules at the edge, then the AWS WAF origin web ACL in BLOCK with the same managed rule groups and route allow-list.
Origin lock	Security group admits only the AWS-managed CloudFront origin-facing prefix list, plus a secret origin header checked by a listener rule or the origin web ACL. The Business and Premium plans also include VPC origins and CloudFront-to-origin mTLS if a stronger lock is wanted. restrict to ALB	Security group admits Imperva's published IP ranges (a list the team maintains), plus mandatory mutual TLS: the ALB trust store validates the client certificate Imperva presents. Imperva-to-origin mTLS
Emergency-page failover	A CloudFront origin group fails GET requests over to the S3 page when the ALB errors or times out. Origin failover is listed only on the Premium plan (USD 1,000) or pay-as-you-go. plan features	Imperva Load Balancing and Site Failover with health monitoring sends traffic to the S3 page. Imperva states the feature's availability depends on the subscription. Imperva failover
Logs into the home-Region archive	CloudFront standard logs and WAF logs to S3; CloudWatch Logs ingestion is included from the Pro plan.	Imperva pushes logs to S3 or SFTP; the feature depends on the subscription, and Imperva buffers logs for at most 48 hours or 500 MB. Imperva log integration
Administrator path	Identical in both: one Verified Access HTTP endpoint for the console, two TCP endpoints for the EKS API and RDS Proxy, IAM Identity Center with mandatory MFA, an AWS WAF web ACL on the Verified Access instance, CloudShell VPC environments for break-glass. Administrators never pass through the public edge in either option. strategy 8.7	
Vendors and residency	AWS only. Cached public content is served from edge locations worldwide, and CloudFront's control plane is a us-east-1 exemption in the Region guardrail (decision B6).	Two vendors. Imperva points of presence and any log data held outside the approved territory are the residency-drift risk R18; the DPA and residency clauses sit in the supplier register.
Worksheet monthly total (defaults, corrected 29 October 2026)	`USD 3,575.70` `Premium plan 1,000 + origin web ACL 21`	`USD 3,579.35` `placeholder 1,000 + origin web ACL 21 + trust store 3.65`

Factor	Proposal 2 · AWS edge	Proposal 3 · Existing edge (Imperva)
Worksheet total before the correction	`USD 2,754.70` `Business plan 200, origin web ACL unpriced`	`USD 3,575.70` `trust store unpriced`
What moves the price	Plan tier against measured traffic. Flat-rate plans have no overage charges, even in a surge or an attack, and origin fetches are free.	The real Imperva quote, its traffic allowance and overage terms, and regional data transfer out of the ALB toward Imperva, which the worksheet excludes.

02 · WHAT THE COMPARISON SHOWS

Five things the totals alone do not say

COST

With failover priced in, the two options cost the same

The worksheet used to start Proposal 2 at the USD 200 Business plan, but that plan has no origin failover and the edge-served emergency page needs one. It now starts at Premium (USD 1,000) plus the origin web ACL, so Proposal 2 lands at USD 3,575.70, USD 3.65 below Proposal 3. The decision turns on the Imperva quote, not on the AWS lines.

OMISSIONS

Each option was missing a small AWS line, now priced

Proposal 2 describes an origin web ACL in BLOCK as a second layer, but the worksheet priced only the CloudFront package: about USD 21 a month at 10 million origin-bound requests. Proposal 3 makes ALB mutual TLS mandatory, and a trust store costs USD 0.005 an hour, about USD 3.65 a month. Both lines are in the worksheet since 29 October 2026.

TRAFFIC

Proposal 3 keeps paying AWS for every byte sent to Imperva

The ALB answers Imperva over the internet, so us-east-1 bills data transfer out at USD 0.09 per GB for the first 10 TB. CloudFront origin fetches are free and the flat-rate plan covers delivery with no overage. This is an existing cost that Proposal 2 removes and the worksheet leaves out of both options.

CONTRACT

Two Imperva features the design relies on are subscription-dependent

Imperva's documentation says both site failover and log integration depend on the subscription. The S3 emergency-page failover (CH-12) and the home-Region log push (B22) only exist in Proposal 3 if the contract includes them. The written scope confirmation that p3-cost asks for must name both.

ADMIN

The administrator path is identical, so it does not decide anything

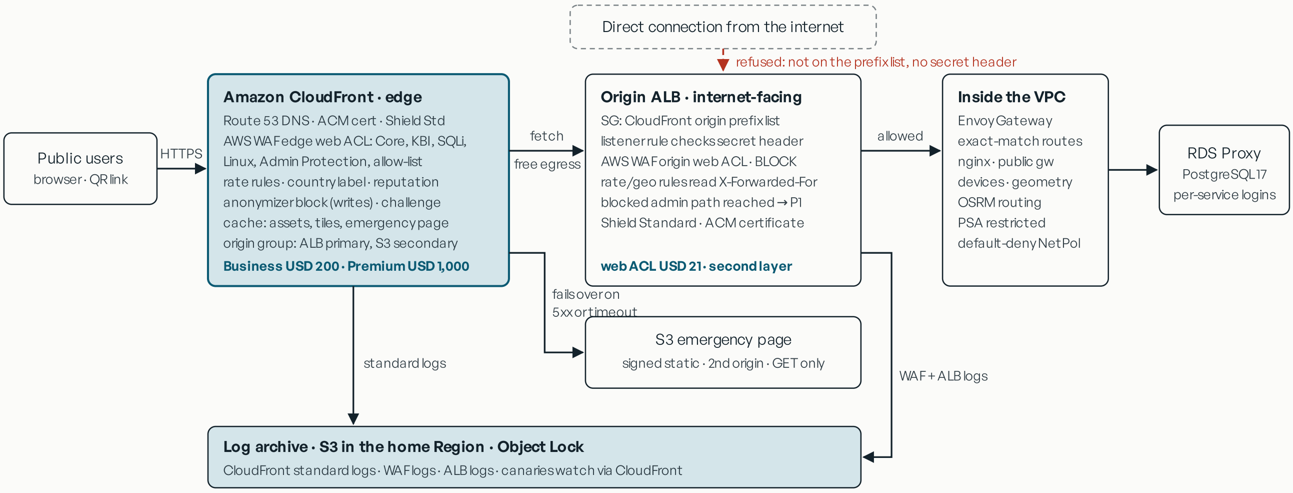
Both options spend USD 504.70 a month on Verified Access and the administrator web ACL, and both keep the console off the public edge. The only difference is which edge the admin plane bypasses. Public and admin planes differ sharply inside each option, which section 06 shows.

03 · FLOW DIAGRAMS

One request through each option, public and admin

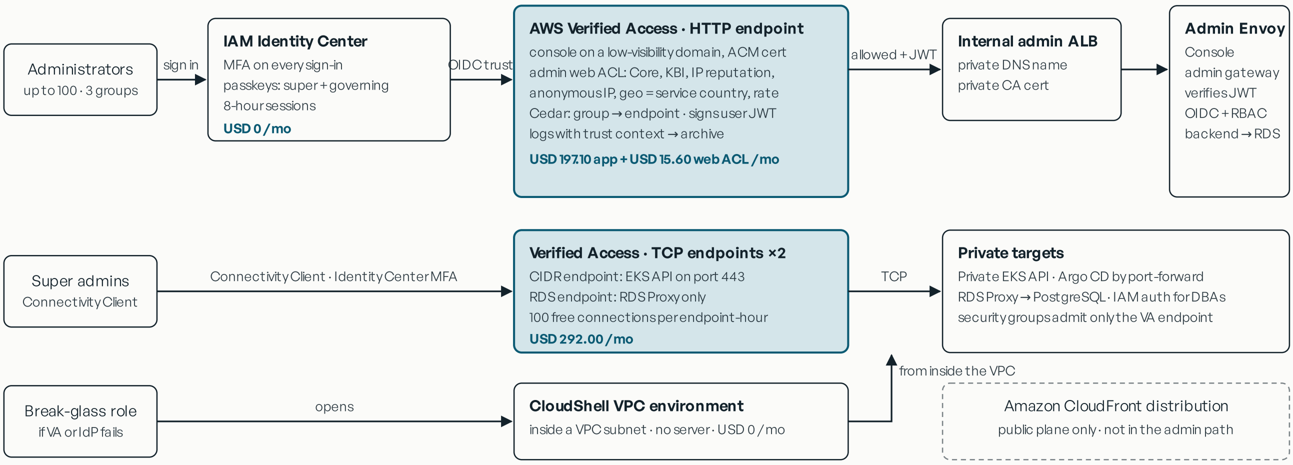
Solid boxes are services the team configures. The tinted box is the public edge the option is choosing. Dashed red is a direct connection that must be refused. Dollar lines are monthly list prices or worksheet placeholders.

Proposal 2 · public path


Direct connection from the internet
refused: not on the prefix list, no secret header
Public users
browser · QR link
HTTPS
Amazon CloudFront · edge
Route 53 DNS · ACM cert · Shield Std
AWS WAF edge web ACL: Core, KBI, SQLi,
Linux, Admin Protection, allow-list
rate rules · country label · reputation
anonymizer block (writes) · challenge
cache: assets, tiles, emergency page
origin group: ALB primary, S3 secondary
Business USD 200 · Premium USD 1,000
fetch
free egress
Origin ALB · internet-facing
SG: CloudFront origin prefix list
listener rule checks secret header
AWS WAF origin web ACL · BLOCK
rate/geo rules read X-Forwarded-For
blocked admin path reached → P1
Shield Standard · ACM certificate
web ACL USD 21 · second layer
allowed
Inside the VPC
Envoy Gateway
exact-match routes
nginx · public gw
devices · geometry
OSRM routing
PSA restricted
default-deny NetPol
RDS Proxy
PostgreSQL 17
per-service logins
fails over on
5xx or timeout
S3 emergency page
signed static · 2nd origin · GET only
standard logs
WAF + ALB logs
Log archive · S3 in the home Region · Object Lock
CloudFront standard logs · WAF logs · ALB logs · canaries watch via CloudFront

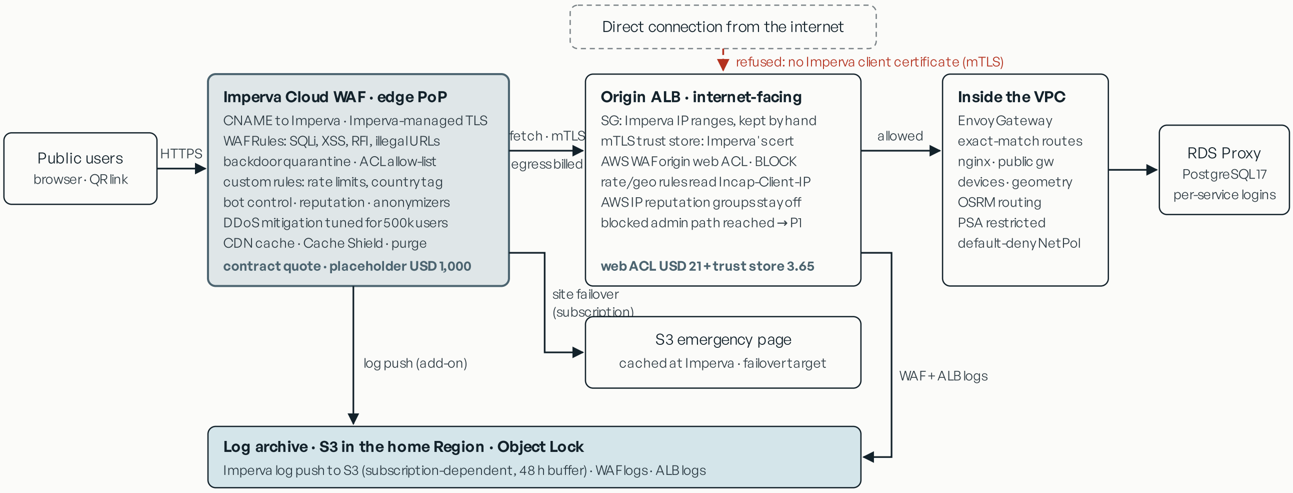
Proposal 2: every public request is filtered and cached at CloudFront before an origin fetch. The origin ALB accepts only the CloudFront prefix list plus the secret header and inspects again with AWS WAF. When the ALB fails, the origin group serves the S3 emergency page for GET requests without a DNS change.

Proposal 2 · administrator path


Administrators
up to 100 · 3 groups
sign in
IAM Identity Center
MFA on every sign-in
passkeys: super + governing
8-hour sessions
USD 0 / mo
OIDC trust
AWS Verified Access · HTTP endpoint
console on a low-visibility domain, ACM cert
admin web ACL: Core, KBI, IP reputation,
anonymous IP, geo = service country, rate
Cedar: group → endpoint · signs user JWT
logs with trust context → archive
USD 197.10 app + USD 15.60 web ACL / mo
allowed + JWT
Internal admin ALB
private DNS name
private CA cert
Admin Envoy
Console
admin gateway
verifies JWT
OIDC + RBAC
backend → RDS
Super admins
Connectivity Client
Connectivity Client · Identity Center MFA
Verified Access · TCP endpoints ×2
CIDR endpoint: EKS API on port 443
RDS endpoint: RDS Proxy only
100 free connections per endpoint-hour
USD 292.00 / mo
TCP
Private targets
Private EKS API · Argo CD by port-forward
RDS Proxy → PostgreSQL · IAM auth for DBAs
security groups admit only the VA endpoint
from inside the VPC
Break-glass role
if VA or IdP fails
opens
CloudShell VPC environment
inside a VPC subnet · no server · USD 0 / mo
Amazon CloudFront distribution
public plane only · not in the admin path

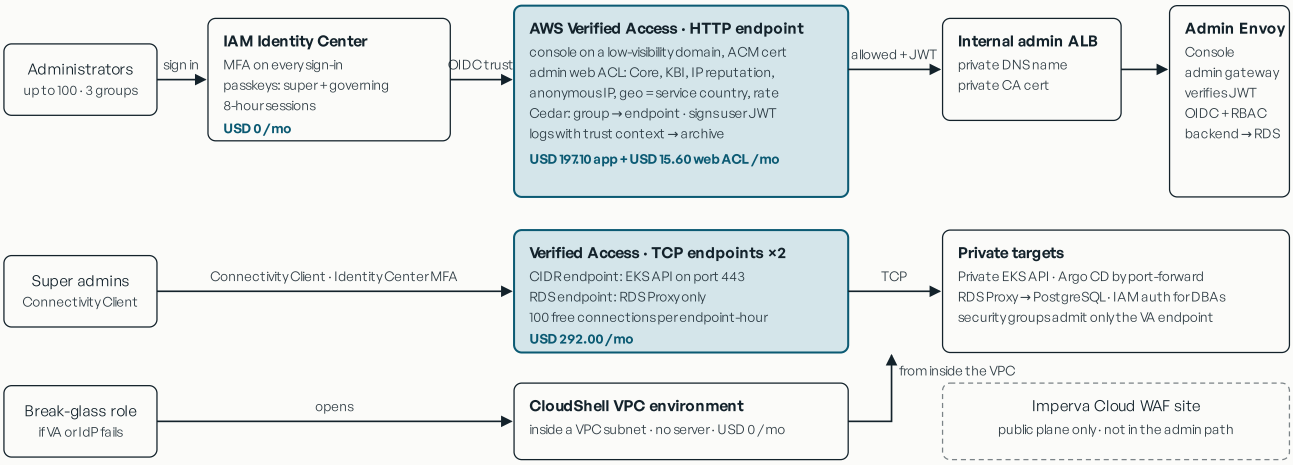
Proposal 2: the console is reached through Verified Access on its own domain, never through the CloudFront distribution. Identity Center enforces MFA, the admin web ACL can use the AWS reputation lists because no proxy sits in front, and the admin gateway verifies the signed user-context token on every request.

Proposal 3 · public path


Direct connection from the internet
refused: no Imperva client certificate (mTLS)
Public users
browser · QR link
HTTPS
Imperva Cloud WAF · edge PoP
CNAME to Imperva · Imperva-managed TLS
WAF Rules: SQLi, XSS, RFI, illegal URLs
backdoor quarantine · ACL allow-list
custom rules: rate limits, country tag
bot control · reputation · anonymizers
DDoS mitigation tuned for 500k users
CDN cache · Cache Shield · purge
contract quote · placeholder USD 1,000
fetch · mTLS
egress billed
Origin ALB · internet-facing
SG: Imperva IP ranges, kept by hand
mTLS trust store: Imperva's cert
AWS WAF origin web ACL · BLOCK
rate/geo rules read Incap-Client-IP
AWS IP reputation groups stay off
blocked admin path reached → P1
web ACL USD 21 + trust store 3.65
allowed
Inside the VPC
Envoy Gateway
exact-match routes
nginx · public gw
devices · geometry
OSRM routing
PSA restricted
default-deny NetPol
RDS Proxy
PostgreSQL 17
per-service logins
site failover
(subscription)
S3 emergency page
cached at Imperva · failover target
log push (add-on)
WAF + ALB logs
Log archive · S3 in the home Region · Object Lock
Imperva log push to S3 (subscription-dependent, 48 h buffer) · WAF logs · ALB logs

Proposal 3: Imperva is the only layer that sees the real visitor, so reputation, anonymizer and country checks happen there. The origin accepts only Imperva's IP ranges with a valid client certificate and inspects again with AWS WAF reading Incap-Client-IP. Failover to the S3 page and the log push to the home Region are Imperva features whose availability depends on the subscription.

Proposal 3 · administrator path


Administrators
up to 100 · 3 groups
sign in
IAM Identity Center
MFA on every sign-in
passkeys: super + governing
8-hour sessions
USD 0 / mo
OIDC trust
AWS Verified Access · HTTP endpoint
console on a low-visibility domain, ACM cert
admin web ACL: Core, KBI, IP reputation,
anonymous IP, geo = service country, rate
Cedar: group → endpoint · signs user JWT
logs with trust context → archive
USD 197.10 app + USD 15.60 web ACL / mo
allowed + JWT
Internal admin ALB
private DNS name
private CA cert
Admin Envoy
Console
admin gateway
verifies JWT
OIDC + RBAC
backend → RDS
Super admins
Connectivity Client
Connectivity Client · Identity Center MFA
Verified Access · TCP endpoints ×2
CIDR endpoint: EKS API on port 443
RDS endpoint: RDS Proxy only
100 free connections per endpoint-hour
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Private targets
Private EKS API · Argo CD by port-forward
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from inside the VPC
Break-glass role
if VA or IdP fails
opens
CloudShell VPC environment
inside a VPC subnet · no server · USD 0 / mo
Imperva Cloud WAF site
public plane only · not in the admin path

Proposal 3: the same administrator mechanism as Proposal 2. Administrators never pass through Imperva, so the two-vendor question does not reach the admin plane, and the admin web ACL's reputation lists work because no proxy is in front.

04 · PUBLIC PATH

Public solution, layer by layer

Layer	Proposal 2 · AWS edge	Proposal 3 · Imperva	What differs
DNS and TLS at the edge	Route 53 record to the distribution; public ACM certificate, free and auto-renewed; both included in every flat-rate plan.	DNS CNAME to the Imperva site; Imperva manages the edge certificate or a custom one is uploaded. The Route 53 hosted zone is billed as today.	Only where the certificate lives. The public domain stays separate from any corporate domain in both (CH-14).
DDoS absorption	Shield Standard on CloudFront and the ALB at no charge. Business and Premium add the Anti-DDoS managed rule group and an uptime SLA. Shield Advanced (USD 3,000 a month, one-year term) is not assumed and is incompatible with flat-rate plans. Shield pricing	Imperva DDoS protection for the site, with the detection threshold tuned for 500,000 concurrent users so a real event is not treated as an attack (WAF-04). The SLA depends on the tier the contract buys (B22). Shield Standard still covers the ALB.	Proposal 3's SLA is contractual; Proposal 2's comes with the Business plan and above.
Application WAF	Edge web ACL with the Core, Known Bad Inputs, SQL database, Linux and Admin Protection managed groups, the positive route allow-list and rate rules. Plans allow 50 (Business) or 75 (Premium) rules including AWS managed rules. Customer rule groups and partner rules are not supported on plans. plan guide	Imperva's default WAF Rules policy blocks SQL injection, cross-site scripting, illegal resource access and remote file inclusion; backdoor protection auto-quarantines; an ACL allow-list and custom rules carry the route list and rate criteria. Imperva policies	Both options keep the AWS managed groups at the origin, so Proposal 3 runs two different engines and Proposal 2 runs AWS WAF twice.
IP reputation and anonymizers	AWS IP reputation and Anonymous IP lists on the edge web ACL, where the request source is the real client; hosting providers and anonymizers blocked on the write path only.	Imperva's reputation and client classification at the edge. The AWS lists stay off at the origin because they would only see Imperva's addresses (WAF-12). WAF-12	Same outcome, different vendor. Proposal 3 is the architecture the strategy was written against.
Bots and challenges	The plan's bot management covers self-identifying (common) bots; targeted bots need pay-as-you-go. JavaScript challenge is included from Business; no CAPTCHA on public pages.	Bot access control denies bad bots and allows good ones by default; a JavaScript challenge only on the position write path and only under attack (WAF-03). Imperva's Advanced Bot Protection is a separate product.	Neither option buys an advanced bot product in release one.
Country checks	Count rule labels by country; stricter rate limits outside the service country; no geo-block on reads; CloudFront geographic restrictions not used.	Same policy at Imperva; hard blocking, if ever wanted, only on the write path through ACL Block Countries.	No difference in policy.

Layer	Proposal 2 · AWS edge	Proposal 3 · Imperva	What differs
Caching	Cache policies control the key; custom caching rules from Business; invalidations included in plans (pay-as-you-go: first 1,000 paths free, then USD 0.005 each). Writes, admin and personal data stay out of the cache. cache key	Imperva cache modes (standard or custom), Cache Shield between PoPs and the origin, and manual purge. Imperva warns that caching HTML can expose personal data, so the same exclusions apply. Imperva cache	Equivalent capability; Proposal 2's publishing procedure uses CloudFront invalidations, Proposal 3's uses Imperva purge.
Origin lock	Security group admits only the AWS-managed CloudFront origin-facing prefix list, and a secret origin header is checked. VPC origins and CloudFront-to-origin mTLS are included from Business if the team wants to take the ALB off the internet entirely.	Security group admits Imperva's published IP ranges, which the team keeps current, and the ALB trust store validates Imperva's client certificate (mandatory mTLS, gate G6). ALB mTLS	Proposal 3 needs a maintained IP list and a trust store (USD 3.65 a month); Proposal 2's list is AWS-managed.
Origin WAF (second layer)	Origin web ACL in BLOCK reads the client from X-Forwarded-For; blocked admin or device-list paths reached at the origin alert at P1. In the worksheet at USD 21.	Origin web ACL in BLOCK reads Incap-Client-IP, which is stripped if it arrives from anywhere else; same P1 alert. In the worksheet at USD 21.	Same control and the same price for both options in the worksheet.
Emergency-page failover	Origin group: ALB primary, S3 page secondary; fails over only after a failed request to the primary and never for POST. Premium plan or pay-as-you-go. origin failover	Imperva health monitoring and site failover to the S3 page, calibrated by the customer; availability depends on the subscription.	Both need a specific tier or add-on; neither is in the cheapest configuration.
Logs and evidence	CloudFront standard logs and WAF logs to the log-archive bucket; CloudWatch Logs ingestion included from Pro. Real-time logs are not supported on plans.	Imperva log integration pushes security and access logs to S3, SFTP or Splunk; subscription-dependent; Imperva keeps at most 48 hours or 500 MB in its repository. Near-real-time SIEM integration is a separate mode.	Proposal 3 must confirm the log feature in the contract and set the S3 push before go-live (B22).
Data transfer	Origin fetches from the ALB to CloudFront are free; viewer delivery is inside the plan allowance (50 TB on Business and Premium) with no overage. pay-as-you-go	The ALB answers Imperva over the internet, so AWS bills Regional data transfer out (USD 0.09 per GB for the first 10 TB at the us-east-1 list price), and Imperva's own bandwidth allowance and overages apply. EC2 data transfer	An existing metered cost that Proposal 2 removes.
Residency	Public content served from edge locations worldwide; CloudFront control plane in us-east-1, listed as an SCP exemption (GP-02); logs delivered to the home Region.	Imperva points of presence and logs outside the approved territory are R18; the DPA, residency commitment and security clauses must be in the supplier register.	Both need a B6 decision; Proposal 3 also needs vendor evidence.

Administrator solution, shared by both options

Component	Mechanism	Monthly USD	Differs between options?
IAM Identity Center	Trust provider with MFA on every sign-in; security keys or passkeys for super and governing-organization administrators; SMS and email codes off; 8-hour sessions (HC-01, HC-07). no extra charge	`0.00`	No
Verified Access HTTP endpoint	Fronts the existing internal administrator ALB on a low-visibility domain with a public ACM certificate; group-based Cedar policies; the admin gateway verifies the signed user-context token (HC-03, HC-08, HC-09). USD 0.27 per hour	`197.10`	No
Administrator web ACL	Attached to the Verified Access instance: Core, Known Bad Inputs, the IP reputation and anonymous IP lists, a geo match for the service country, a per-IP rate limit and an admin route allow-list (HC-04, HC-05). One ACL, ten rule entries, one million requests.	`15.60`	No. The reputation lists work in both because no proxy sits in front.
Verified Access TCP endpoints	Network CIDR endpoint for the EKS API on port 443 and an RDS endpoint targeting RDS Proxy, used through the Connectivity Client on Windows and macOS; 100 free connections per endpoint-hour (HC-10). USD 0.20 per hour each	`292.00`	No
CloudShell VPC environments	Break-glass and Linux path when Verified Access or the identity provider fails; reaches the EKS API, RDS Proxy and the admin gateway from inside the VPC (HC-12).	`0.00`	No
Internal admin ALB, admin Envoy, console, admin gateway, backend	Existing components; private DNS name, private CA certificate; application OIDC login and role-based authorization still enforced (R01).	`existing`	No
Client VPN	Runs in parallel for two weeks with a written rollback, then the endpoint is removed (CH-15).	`removed`	No
Admin plane, priced lines	**Same services, same configuration, same evidence in both options.**	`504.70`	**Only the edge it bypasses: CloudFront in Proposal 2, Imperva in Proposal 3.**

Cost basis: 730 hours a month and the rates on the Verified Access and AWS WAF pricing pages, as the worksheet assumes. Verified Access data processing at USD 0.02 per GB sits inside the Common usage allowance.

06 · PUBLIC VS ADMIN, PER OPTION

Inside each option, how the two planes differ

Proposal 2 · AWS edge

Aspect	Public plane	Admin plane
Entry point	CloudFront distribution on the public domain, reachable by anyone.	Verified Access HTTP endpoint on a separate low-visibility domain; the console stays entirely off the distribution.
Identity	Anonymous visitors; device checks and PostGIS validation happen in the application.	Identity Center with mandatory MFA, Cedar group policies, then the application's own OIDC login and RBAC.
Web firewall	Edge web ACL in the plan (managed rules, allow-list, rate, country label, reputation) plus the origin web ACL in BLOCK.	One administrator web ACL on the Verified Access instance with reputation lists, geo match and a short admin route allow-list.
Location	Labelled, never blocked on reads; stricter rate limits abroad; anonymizers blocked on writes.	Allowed only from the service country, enforced at the identity provider and in Cedar, with a time-boxed exception ticket.
Caching and challenges	Assets, tiles and the emergency page cached; JavaScript challenge available on the write path.	Nothing cached; no challenge; every request re-evaluated against the policy.
Load balancer	Internet-facing origin ALB locked to the CloudFront prefix list and secret header.	Internal admin ALB with a private name and private CA certificate; security groups admit only the Verified Access endpoint.
Failure path	Origin group serves the S3 emergency page for GET requests.	Break-glass role opens a CloudShell VPC environment.
Logs and alerts	CloudFront and WAF logs; P1 when a blocked path reaches the origin; canaries through CloudFront.	Verified Access logs with trust context; P1 on policy, trust-provider or web ACL changes, MFA resets and out-of-inventory administrators; P2 on unexpected countries or hours.
Monthly USD (list or worksheet)	`1,000.00 Premium (200.00 Business has no failover) + origin web ACL 21.00`	`504.70`

Proposal 3 · Existing edge

Aspect	Public plane	Admin plane
Entry point	Imperva site on the public domain; Imperva terminates TLS and proxies to the origin.	Verified Access HTTP endpoint on a separate domain; administrators never pass through Imperva.
Identity	Anonymous visitors; application-side validation of every coordinate.	Identity Center with mandatory MFA, Cedar policies, application OIDC and RBAC.
Web firewall	Imperva WAF Rules policy, ACL and custom rules at the edge, then the AWS WAF origin web ACL in BLOCK.	One administrator web ACL on the Verified Access instance.

Aspect	Public plane	Admin plane
Who sees the visitor	Imperva only; the origin reads Incap-Client-IP and strips it from any other source.	Verified Access sees the administrator directly, so the AWS reputation lists work here.
Load balancer	Internet-facing origin ALB admitting Imperva IP ranges with mandatory mutual TLS (trust store).	Internal admin ALB, private name, private CA; only the Verified Access endpoint admitted.
Failure path	Imperva site failover to the S3 emergency page (subscription-dependent).	Break-glass role opens CloudShell.
Logs and alerts	Imperva push to S3 (subscription-dependent) plus WAF and ALB logs; Imperva alerts are P2 and P1 during an event (AL-10).	Verified Access logs with trust context; same P1 and P2 rules as Proposal 2.
Vendors	Two: Imperva and AWS, coordinated in one response process and one runbook per provider.	One: AWS.
Monthly USD (list or worksheet)	`Imperva quote (placeholder 1,000.00) + origin web ACL 21.00 + trust store 3.65 + Regional data transfer (existing, excluded from the worksheet)`	`504.70`

Each AWS service against what Imperva includes

"Included" means Imperva documents the capability as part of Cloud WAF. "Subscription-dependent" means Imperva's documentation says availability depends on the plan, so the contract must name it. "Not included" means AWS provides it in both options or Imperva has no equivalent in the path.

AWS service or feature	Role in Proposal 2	Role in Proposal 3	Imperva includes it?	Cost effect
Amazon CloudFront (CDN, TLS, caching, invalidations)	Public entry and cache; one distribution on a flat-rate plan.	Not used. Imperva's CDN caches, with Cache Shield and purge.	Included	Proposal 2: USD 200 (Business) or USD 1,000 (Premium), or pay-as-you-go. Proposal 3: inside the Imperva contract; no AWS line.
AWS WAF web ACL at the edge	On CloudFront: managed groups, allow-list, rate rules, country labels. Mandatory on a plan; counts against 50 or 75 rules.	None. Imperva WAF Rules policy, ACL and custom rules do this at the edge.	Included	Proposal 2: bundled in the plan (pay-as-you-go would be USD 5 + rules + USD 0.60 per million). Proposal 3: contract.
AWS WAF web ACL at the origin ALB	Second layer in BLOCK; reads X-Forwarded-For.	Second layer in BLOCK; reads Incap-Client-IP.	Not applicable	Both: USD 5 + USD 10 for ten rule entries + USD 0.60 per million origin-bound requests, about USD 21 at the worksheet volume. Priced for both options in the worksheet.
AWS managed rule groups (Core, Known Bad Inputs, SQL database, Linux, Admin Protection)	At the edge and at the origin.	At the origin only.	Partly Imperva's policy covers SQLi, XSS, RFI and illegal resource access; the Java deserialization and Log4j rules in Known Bad Inputs are AWS-specific.	USD 1 per managed group per web ACL, inside the ten-entry assumption. No change between options.
AWS WAF IP reputation and Anonymous IP lists	On the edge web ACL, effective.	Off on the public plane; on the admin web ACL only.	Included Imperva reputation and client classification.	No price difference; two managed groups either way.
AWS WAF Bot Control	Common bots are part of Business and Premium; targeted bots are not supported on plans.	Not used; Imperva bot access control.	Included Basic bot control. Advanced Bot Protection is a separate Imperva product.	USD 0 in both as designed. Targeted Bot Control would be USD 10 a month plus USD 10 per million on pay-as-you-go. WAF pricing

AWS service or feature	Role in Proposal 2	Role in Proposal 3	Imperva includes it?	Cost effect
AWS WAF JavaScript challenge and CAPTCHA	JS challenge in Business and Premium; CAPTCHA responses in rules included in plans.	Not used; Imperva challenge on the write path under attack.	Included	Pay-as-you-go would bill USD 0.40 per million challenge responses; nothing inside a plan or inside Imperva.
AWS Shield Standard	CloudFront and the ALB.	The ALB behind Imperva.	Included Imperva DDoS mitigation for the site; threshold and SLA per tier.	USD 0 in both.
AWS Shield Advanced	Not assumed; not allowed on a flat-rate distribution.	Not assumed.	Depends on tier Imperva's DDoS SLA is the counterpart (B22).	USD 3,000 a month with a one-year term if ever bought; neither option includes it.
AWS WAF Anti-DDoS managed rule group	Included from Business ("Advanced DDoS protection").	Not used.	Included Layer 7 DDoS mitigation is Imperva's core service.	Pay-as-you-go would be USD 20 per web ACL plus USD 0.15 per million.
CloudFront origin group (origin failover)	ALB primary, S3 secondary, GET only.	Not used; Imperva Load Balancing and Site Failover with health monitoring.	Subscription-dependent	Proposal 2: Premium plan (USD 1,000) or pay-as-you-go, where failover has no separate charge. Proposal 3: confirm it is in the contract.
Amazon S3 (signed static emergency page)	Secondary origin behind the origin group.	Failover target cached at Imperva.	Not applicable	Negligible; Business includes 1 TB of S3 storage credits.
CloudFront origin-facing managed prefix list and secret origin header	Locks the ALB to CloudFront.	Replaced by Imperva's published IP ranges, maintained by the team.	Partly Imperva publishes its IP ranges; nothing updates the security group automatically.	USD 0 in both; Proposal 3 carries an operational task.
CloudFront VPC origins and CloudFront-to-origin mTLS	Available from Business if the ALB should leave the internet entirely; not required by the proposal text.	Not applicable.	Included Imperva-to-origin mTLS is the Proposal 3 equivalent.	Inside the plan. A design choice, not a cost line.
ALB mutual TLS trust store	Not needed.	Mandatory: validates Imperva's client certificate (G6).	Included Imperva uploads and presents the client certificate.	USD 0.005 per trust-store hour, about USD 3.65 a month; in the worksheet. ELB pricing
AWS WAF forwarded-IP configuration	Origin rules read X-Forwarded-For.	Origin rules read Incap-Client-IP; header stripped from any other source.	Included Imperva inserts the header by default.	USD 0. forwarded IP

AWS service or feature	Role in Proposal 2	Role in Proposal 3	Imperva includes it?	Cost effect
Amazon Route 53	Public DNS for the distribution, included in the plan.	CNAME to Imperva from the existing hosted zone.	Separate product Imperva DNS Protection is a different service, not assumed.	Hosted zone and queries are existing costs in both.
AWS Certificate Manager	Public certificate on CloudFront and on the origin ALB.	Certificate on the origin ALB; Imperva handles the edge certificate.	Included	USD 0 in both.
CloudFront standard logs and WAF logs to S3	Delivered to the home-Region archive; CloudWatch Logs ingestion included from Pro.	WAF and ALB logs as usual; edge logs come from Imperva's push.	Subscription-dependent Imperva log integration to S3, SFTP or Splunk; 48-hour buffer at Imperva.	S3 storage sits in the Common allowance for both. Proposal 3 must confirm the log feature in the contract.
CloudWatch Synthetics canaries	Watch the site and emergency page through CloudFront.	Watch them through Imperva.	Not applicable	Same in both; inside the Common monitoring allowance.
AWS Verified Access (one HTTP app, two TCP endpoints)	Administrator path.	Administrator path.	Not included Imperva plays no part in administrator access.	USD 489.10 a month in both.
IAM Identity Center and CloudShell VPC environments	Trust provider and break-glass.	Same.	Not included	USD 0 in both.
AWS WAF web ACL on the Verified Access instance	Administrator web ACL.	Same.	Not included	USD 15.60 a month in both.
Regional data transfer out (ALB to the internet)	Not billed: origin fetches to CloudFront are free, delivery is inside the plan.	Billed for every byte the ALB sends to Imperva, at USD 0.09 per GB for the first 10 TB (us-east-1 list price).	Partly Imperva's bandwidth allowance covers its own delivery, not the AWS egress toward it.	Proposal 3 continues an existing metered cost, reduced only by what Imperva caches. Excluded from the worksheet in both, and the worksheet now says so.
Network Firewall, VPC endpoints, EKS, RDS Proxy, KMS, Secrets Manager, GuardDuty, Security Hub, Inspector, Config, Backup	Identical inside the VPC and in Common.	Identical.	Not included	No difference; Common allowance or existing hosting.

Where the money goes

Worksheet defaults, side by side (corrected 29 October 2026)

Line	Proposal 2	Proposal 3	Basis
Common (detection, logs and backups, operations, licences)	2,050.00	2,050.00	Planning allowances, identical.
Verified Access (one web application, two TCP endpoints)	489.10	489.10	730 hours at USD 0.27 and 2 × USD 0.20.
Administrator web ACL	15.60	15.60	USD 5 + 10 rule entries + 1 million requests.
Public edge	1,000.00	1,000.00	CloudFront Premium plan example, the lowest flat-rate plan that lists origin failover, versus the Imperva placeholder, which is not a quote.
AWS WAF at the origin	21.00	21.00	USD 15 + USD 0.60 × 10 million, counting every public request as origin-bound.
ALB mutual TLS trust store	–	3.65	730 hours × USD 0.005; mandatory in Proposal 3 only.
Monthly	3,575.70	3,579.35	**USD 3.65 apart. The Imperva placeholder decides the rest.**
Yearly recurring	42,908.40	42,952.20	Monthly × 12.
One-time (shared setup 8,000 + option setup 3,000)	11,000.00	11,000.00	Both options assume 30 hours of implementation.
First year	53,908.40	53,952.20	Yearly plus one-time.

What the correction changed

Line	Proposal 2	Proposal 3	Why
Worksheet monthly before 29 October 2026	2,754.70	3,575.70	Business plan at USD 200 and no origin web ACL for Proposal 2; no trust store for Proposal 3.
Edge-served failover needs Premium	+ 800.00	–	Origin failover is listed only on Premium (USD 1,000); Business (USD 200) lacks it. p2-cost now says so.
Origin web ACL in BLOCK	+ 21.00	–	p2-edge keeps the origin web ACL as a second layer; the worksheet now prices it.
ALB mutual TLS trust store	–	+ 3.65	p3-origin makes mTLS mandatory; USD 0.005 per hour per trust store, now a worksheet line.
Corrected monthly (the current default)	3,575.70	3,579.35	**USD 3.65 apart, as above.**
Corrected first year	53,908.40	53,952.20	Both with USD 11,000 one-time.

Line	Proposal 2	Proposal 3	Why
Pay-as-you-go alternative for Proposal 2 at the worksheet volume	2,596.70	—	10 million requests and 1 TB fall inside CloudFront's always-free tier, so the edge costs only two web ACLs (USD 42). Origin failover has no separate charge on pay-as-you-go, but the plan's Anti-DDoS rules, uptime SLA and no-overage guarantee are not part of it.

How the public edge scales with traffic

Public-edge lines only, excluding Common, the admin plane and the Imperva quote. Volumes are illustrations, not forecasts: data assumes an average of 100 KB per request. Proposal 2 pay-as-you-go uses the Europe price class and charges the first-10 TB rate for all volume beyond the free 1 TB, an upper bound. Proposal 3 uses the us-east-1 data transfer tiers, and Imperva caching reduces origin-bound volume.

Requests and data per month	P2 pay-as-you-go	P2 Business (no failover)	P2 Premium	P3 AWS side	P3 with USD 1,000 placeholder
10 million · 1 TB (worksheet default)	21.00	200.00	1,000.00	114.65	1,114.65
50 million · 5 TB	433.00	200.00	1,000.00	498.65	1,498.65
125 million · 12.5 TB (Business allowance)	1,205.50	200.00	1,000.00	1,206.15	2,206.15
300 million · 30 TB (event month)	3,008.00	above allowance	1,000.00	2,798.65	3,798.65

P2 pay-as-you-go = CloudFront data transfer (USD 0.085 per GB after 1 TB free) + HTTPS requests (USD 0.012 per 10,000 after 10 million free) + edge web ACL (USD 15 + USD 0.60 per million). P3 AWS side = origin web ACL (USD 15 + USD 0.60 per million) + Regional data transfer out (USD 0.09 per GB first 10 TB, USD 0.085 next 40 TB) + trust store (USD 3.65). Add the origin web ACL to every P2 column for a like-for-like total. The flat-rate allowances are not hard limits, but AWS notifies at 50, 80 and 100 percent and the plan is sized for that baseline; blocked requests never count. Premium can be configured up to 6 billion requests and 600 TB for a new flat rate.

What the worksheet leaves out, and which way each item leans

Item	Proposal 2	Proposal 3	Leans toward
Regional data transfer out from the ALB	Removed: origin fetches to CloudFront are free.	Continues: every byte to Imperva is billed by the Region; the worksheet now names this exclusion.	Proposal 2
Surge and attack months	No overage on a flat-rate plan; blocked requests do not count.	Imperva traffic allowances and overage terms apply; AWS egress is metered.	Proposal 2

Item	Proposal 2	Proposal 3	Leans toward
Migration	Build and test a new edge, origin lock, caching and failover; run the origin-bypass and cache-update tests.	No migration; the existing site stays and the team adds mTLS, the origin web ACL and the log push.	Proposal 3
Incident support and DDoS SLA	AWS uptime SLA from Business; Shield Response Team only with Shield Advanced and a Business or Enterprise support plan.	Contractual DDoS SLA and 24/7 Imperva security operations on the right tier.	Proposal 3, if the contract has it
Site failover and log push	Failover on Premium or pay-as-you-go; logs from Pro.	Both subscription-dependent at Imperva; must be in the written scope.	Unknown until the quote
Operational coordination	One vendor; the team tunes cache and invalidation rules and keeps the origin header secret.	Two vendors; WAF rules, log delivery, residency evidence and incidents coordinated across AWS and Imperva; the IP range list maintained by hand.	Proposal 2
Residency evidence	Edge locations worldwide and a us-east-1 control plane need a B6 statement.	Imperva DPA, residency commitment and log location need supplier evidence (R18).	Neither; both need a steering-committee decision
Shield Advanced, targeted Bot Control, fraud control	Not assumed; not available on a flat-rate plan.	Not assumed.	Neither

09 · BEFORE APPROVAL

What each option must prove, and what the quote must say

PROPOSAL 2

Approve only after

A traffic model with normal and emergency forecasts; an origin-bypass test proving a direct connection to the ALB is refused (G6); a cache-update test against the publishing procedure; the surge drill with edge failover and an emergency guidance update (G11); a residency review accepting worldwide edge delivery; and a decision between Premium and pay-as-you-go once the failover requirement is priced.

PROPOSAL 3

Approve only after

A written Imperva price and scope confirmation naming the tier, bot management, the DDoS SLA, site failover, log integration to S3, traffic allowances and overage terms; mandatory origin mutual TLS tested (G6); the log push to the home Region confirmed before go-live (B22); the DPA and residency clauses in the supplier register; and the comparison of that total plus the avoided migration against Proposal 2.

BOTH

Unchanged either way

Verified Access with Identity Center MFA for administrators, the origin web ACL in BLOCK after a clean seven-day COUNT period, exact-match public routes, the signed S3 emergency page, Kyverno in Enforce, and the twelve go-live gates with evidence.

> If the Imperva quote lands near the USD 1,000 placeholder, the two options cost the same on the AWS lines and the choice is about operations and contract value: one vendor with no overage and no migration risk on AWS egress, against an existing edge with a contractual DDoS SLA and no migration. If the quote is materially higher, Proposal 2 on Premium is cheaper and also removes the metered egress. If it is materially lower, Proposal 3 wins on price but still needs failover and log push in the contract.

Documents this comparison relies on

Project documents

1. Management proposal, Proposal 2 items p2-fit to p2-work and Proposal 3 items p3-fit to p3-work, and the budget worksheet: management proposal
2. Security strategy 8.1 WAF rules (WAF-01 to WAF-14), 8.5 admin hostname, 8.7 admin access (HC-01 to HC-12), decisions B6 and B22, risks R05, R07 and R18, changes CH-12 and CH-15, gates G6 and G11: security strategy

AWS, checked 29 October 2026

1. CloudFront pricing, flat-rate plans: aws.amazon.com/cloudfront/pricing/
2. CloudFront flat-rate pricing plan guide, features by tier, usage allowances, unsupported features: docs.aws.amazon.com/AmazonCloudFront/latest/DeveloperGuide/flat-rate-pricing-plan.html
3. CloudFront pay-as-you-go pricing: aws.amazon.com/cloudfront/pricing/pay-as-you-go/
4. AWS WAF pricing: aws.amazon.com/waf/pricing/
5. AWS Verified Access pricing: aws.amazon.com/verified-access/pricing/
6. AWS Shield pricing: aws.amazon.com/shield/pricing/
7. Elastic Load Balancing pricing, including the mutual TLS trust-store rate: aws.amazon.com/elasticloadbalancing/pricing/
8. EC2 on-demand pricing, data transfer out to the internet: aws.amazon.com/ec2/pricing/on-demand/
9. CloudFront origin failover: high_availability_origin_failover.html
10. Restricting access to an Application Load Balancer origin: restrict-access-to-load-balancer.html
11. CloudFront VPC origins: private-content-vpc-origins.html
12. AWS WAF IP reputation rule groups: aws-managed-rule-groups-ip-rep.html
13. AWS WAF forwarded IP address: waf-rule-statement-forwarded-ip-address.html
14. Application Load Balancer mutual TLS: mutual-authentication.html
15. IAM Identity Center FAQ (no additional charge): aws.amazon.com/iam/identity-center/faqs/

Imperva, checked 29 October 2026

1. Imperva Cloud WAF datasheet (CDN, DDoS Protection, Advanced Bot Protection, API Security and Load Balancing as a suite): cloud-web-application-firewall.pdf
2. Create and manage policies (WAF Rules, ACL): policies.htm
3. WAF settings and security settings: waf-settings.htm, security-settings.htm
4. Rules (rate criteria): rules.htm
5. Cache settings and Cache Shield: caching-settings.htm
6. Load Balancing and Failover (availability depends on subscription): load-balancing-failover.htm
7. Cloud WAF Log Integration (availability depends on subscription; push to SFTP, Amazon S3 or Splunk; 48-hour repository): log-integration.htm
8. Imperva-to-Origin mTLS: certificate-mtls.htm
9. Customer setup checklist (Incap-Client-IP and X-Forwarded-For headers): setup-checklist.htm

Every amount is a list price in US dollars at the Region shown on the pricing page, or a worksheet planning placeholder. None is a vendor quote, a capacity model or a promise of savings. Confirm us-east-1 rates, Verified

Access availability and the Imperva contract before approval.